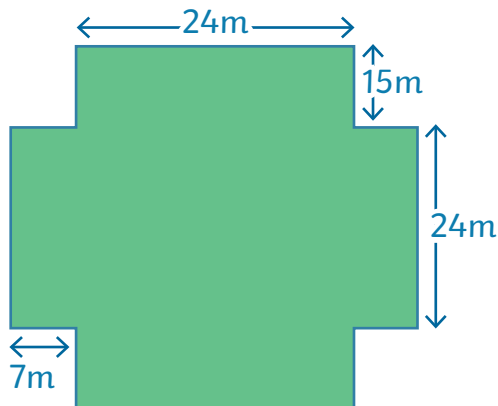


Perimeter, Area and Volume: At the Farm

Farmer Kiki has a farm. She wants to develop it but needs some help with the calculations. Read each problem carefully and calculate the answers.

1

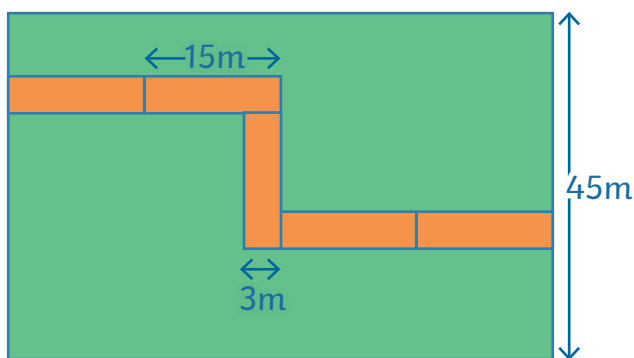
Kiki is replacing the wire fencing on her symmetrical chicken enclosure. Each 10m roll of wire costs £27.84. This is a plan of her enclosure.



How much will it cost to put wire around the perimeter?

2

Kiki is creating a path through a field she is sowing carrots seeds into. She will use 5 identical wooden boards to make her path.



a) What is the area of the space that she can sow her seeds once the path is completed?

b) What is the perimeter of the path?

3

Kiki buys some boxes of grass seed. Here is a table showing the measurements of her boxes.

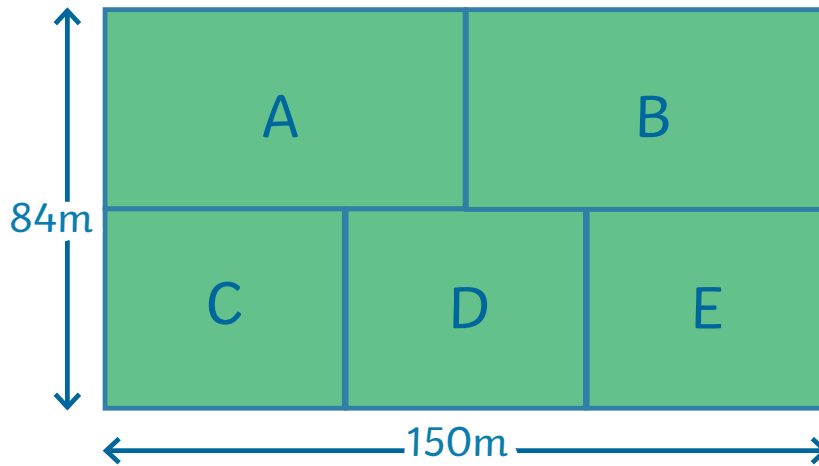
a) Fill in the missing information.

Product	Box Length	Box Width	Box Height	Volume of Box
Farmer's Delight	40cm	21cm	15cm	
Super Seeds		24cm	26cm	5616cm ³
True Lawn	32cm		17cm	5984cm ³
Green and Lovely	29cm	18cm		7308cm ³
Fast Gro (comes in a cube-shaped box)				1728cm ³

b) The True Lawn package comes wrapped in paper. What area of paper is required to cover all of the faces?

4

Kiki decides to remove some fence panels which currently divide a field in order to make a bigger field. Here is a plan of the current fields.



Fields A and B are the same size. Fields C, D and E are the same size.

a) If Kiki removes the fence between field A and field B, what will the area of the new field be?

b) If Kiki removes the fence between field B and field D, what length of fence will she need to remove?

c) If Kiki joins fields A, C and D to make a new field, what will the perimeter be of this new field?

d) If Kiki makes fields A, B and D into one big field, what will the area be of this new field?

e) Kiki decides to remove two lengths of fencing to join field A to field C and then join field B to field E. She says, "The total area of my two new fields is five times the area of field D."

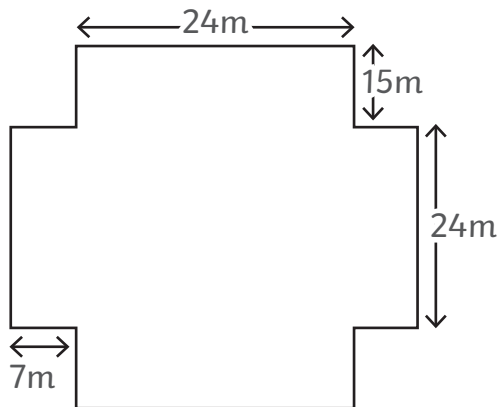
Is she correct?

Perimeter, Area and Volume: At the Farm Answers

Farmer Kiki has a farm. She wants to develop it but needs some help with the calculations.

Read each problem carefully and calculate the answers.

1. Kiki is replacing the wire fencing on her symmetrical chicken enclosure. Each 10m roll of wire costs £27.84. This is a plan of her enclosure.



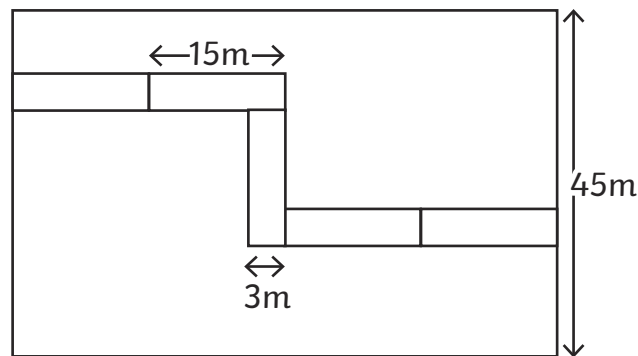
How much will it cost to put wire around the perimeter?

$$\begin{aligned}\text{perimeter} &= 24\text{m} + 15\text{m} + 7\text{m} + 24\text{m} + 7\text{m} + \\ &15\text{m} + 24\text{m} + 15\text{m} + 7\text{m} + 24\text{m} + 7\text{m} + 15\text{m} \\ &= 96\text{m} + 60\text{m} + 28\text{m} \\ &= 184\text{m}\end{aligned}$$

$184 \div 10 = 18.4$ rolls of wire needed so will need to buy 19 rolls:

$$19 \times £27.84 = £528.96$$

2. Kiki is creating a path through a field she is sowing carrots seeds into. She will use 5 identical wooden boards to make her path.



a) What is the area of the space that she can sow her seeds once the path is completed?

$$\text{area of field including path} = 45\text{m} \times 60\text{m} = 2700\text{m}^2$$

$$\text{area of one wooden board} = 15\text{m} \times 3\text{m} = 45\text{m}^2$$

$$5 \text{ wooden board used so area} = 5 \times 45\text{m}^2 = 225\text{m}^2$$

$$\text{total area of field} - \text{area of path} = 2700\text{m}^2 - 225\text{m}^2 = 2475\text{m}^2$$

b) What is the perimeter of the path?

$$15\text{m} + 15\text{m} + 3\text{m} + 12\text{m} + 15\text{m} + 15\text{m} + 3\text{m} + 15\text{m} + 15\text{m} + 3\text{m} + 15\text{m} + 12\text{m} + 15\text{m} + 3\text{m}$$

$$= 120 + 12\text{m} + 24\text{m}$$

$$= 156\text{m}$$

3. Kiki buys some boxes of grass seed. Here is a table showing the measurements of her boxes.

a) Fill in the missing information.

Product	Box Length	Box Width	Box Height	Volume of Box
Farmer's Delight	40cm	21cm	15cm	12 600cm³
Super Seeds	9cm	24cm	26cm	5616cm ³
True Lawn	32cm	11cm	17cm	5984cm ³
Green and Lovely	29cm	18cm	14cm	7308cm ³
Fast Gro (comes in a cube-shaped box)	12cm	12cm	12cm	1728cm ³

b) The True Lawn package comes wrapped in paper. What area of paper is required to cover all of the faces?

True Lawn measurements:

length 32cm

width 11cm

height 17cm

2 faces are 32cm x 11cm:

$$32\text{cm} \times 11\text{cm} = 352\text{cm}^2$$

$$352\text{cm}^2 \times 2 = 704\text{cm}^2$$

2 faces are 11cm x 17cm:

$$11\text{cm} \times 17\text{cm} = 187\text{cm}^2$$

$$187\text{cm}^2 \times 2 = 374\text{cm}^2$$

2 faces are 32cm x 17cm:

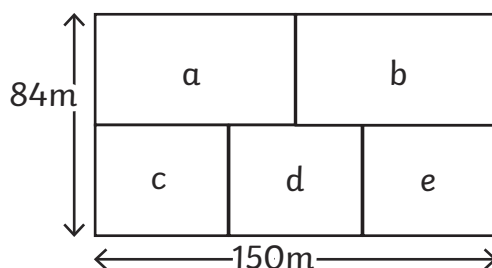
$$32\text{cm} \times 17\text{cm} = 544\text{cm}^2$$

$$544\text{cm}^2 \times 2 = 1088\text{cm}^2$$

$$\text{total area} = 704\text{cm}^2 + 374\text{cm}^2 + 1088\text{cm}^2$$

$$= 2166\text{cm}^2$$

4. Kiki decides to remove some fence panels which currently divide a field in order to make a bigger field. Here is a plan of the current fields.



Fields A and B are the same size. Fields C, D and E are the same size.

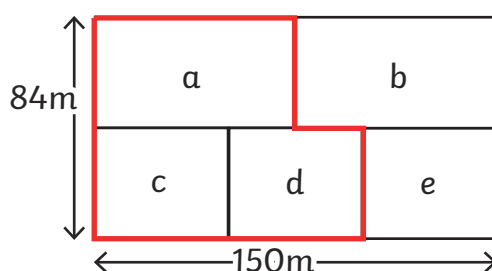
a) If Kiki removes the fence between field A and field B, what will the area of the new field be?

$$150\text{m} \times 42\text{m} = 6300\text{m}^2$$

b) If Kiki removes the fence between field B and field D, what length of fence will she need to remove?

$$150\text{m} \div 3 \div 2 = 50\text{m} \div 2 = 25\text{m}$$

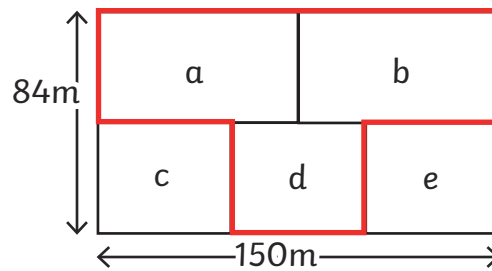
c) If Kiki joins fields A, C and D to make a new field, what will the perimeter be of this new field? Here is the plan with the new field outlined in red.



$$\begin{aligned}\text{perimeter} &= 75\text{m} + 42\text{m} + 25\text{m} + 42\text{m} + 100\text{m} + 84\text{m} \\ &= 168\text{m} + 200\text{m} \\ &= 368\text{m}\end{aligned}$$

d) If Kiki makes fields A, B and D into one big field, what will the area be of this new field?

Here is the plan with the new field outlined in red:



area of A = $75\text{m} \times 42\text{m} = 3150 \text{ m}^2$

area of B = $75\text{m} \times 42\text{m} = 3150\text{m}^2$

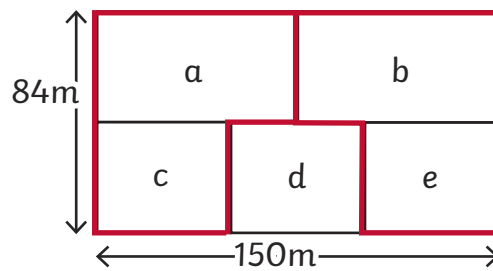
area of C = $50\text{m} \times 42\text{m} = 2100\text{m}^2$

total area = $3150\text{m}^2 + 3150\text{m}^2 + 2100\text{m}^2 = 8400\text{m}^2$

e) Kiki decides to remove two lengths of fencing to join field A to field C and then join field B to field E. She says, "The total area of my two new fields is five times the area of field D."

Is she correct?

Here is the plan with the new fields outlined in red:



area of A = $75\text{m} \times 42\text{m} = 3150\text{m}^2$

area of C = $50\text{m} \times 42\text{m} = 2100\text{m}^2$

total area of A and C = $3150\text{m}^2 + 2100\text{m}^2 = 5250\text{m}^2$

The area of B and E will be the same as the area of A and C so the total area of the two new fields = $2 \times 5250\text{m}^2 = 10\,500\text{m}^2$

Area of D = $50\text{m} \times 42\text{m} = 2100\text{m}^2$

$2100\text{m}^2 \times 5 = 10\,500\text{m}^2$ so Kiki is correct.